

name: _____

openhouse

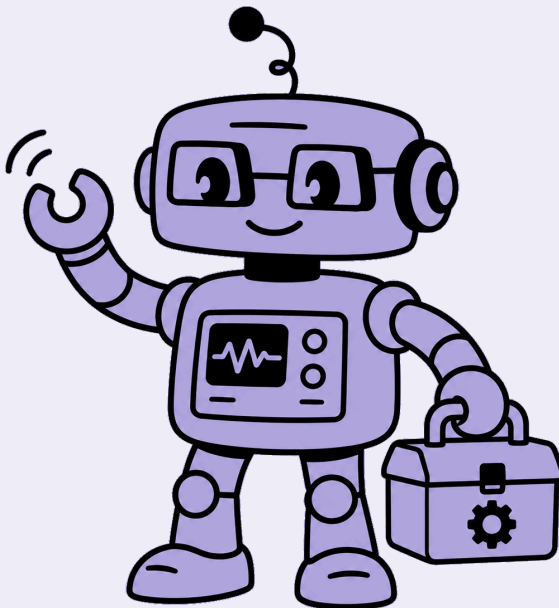
robotics • level 1

◆ mechanics ◆

your portfolio of the openhouse experience

ages 8-12

part 1 • machines 1-5



why mechanics?

This is an **engineering** book. Engineers build machines and work out how they work.

Machines are everywhere – a see-saw, a crane, a bicycle. **Mechanics** is how the simple machines inside them work: levers, pulleys, gears and wheels. When you learn it, you can build your own machines and say why they work.

You grow four **skills** and build ten real **machines**. For each one you explore it, run an experiment, build it, solve a problem, and show it. The next pages show how.

the skills you build

You grow all four skills on **every machine**. Each grows in four **outcomes** (the steps below) – the machines climb them in order, a little further each time. You are on outcomes 1–2 in this book.



observing & understanding

1. measure what the machine does →
 2. record the pattern in numbers →
 3. compare and work out the rule →
 4. **predict a value, then prove it**
-



building & making

1. build it and read the parts →
 2. build it and label the working parts →
 3. build a machine with several parts →
 4. **improve it and measure the change**
-



problem solving

1. find what isn't working →
 2. predict where the fault is →
 3. change one thing and measure the effect →
 4. **solve it and explain why it worked**
-



presenting & explaining

1. name the machine and its parts →
 2. explain how it works in the right words →
 3. explain why, using your result →
 4. **present it and answer a question**
-

the machines

the machines you build

Ten real machines across two books. Each uses a simple machine – a lever, a pulley, gears, or a wheel and axle.

part 1 · machines 1-5

see-saw	uses a lever
weighing scale	uses a lever
crane	uses a pulley
wind turbine	uses a pulley
copter	uses gears

part 2 · machines 6-10

bulldozer	uses gears
rickshaw	uses a wheel and axle
fishing rod	uses two machines in one
drawbridge	uses two machines in one
tow truck	uses two machines in one

what's inside this book

For **every machine** you do five things – you can do them on your own:

1 explore
a story, a quick activity, and the words you'll use

2 experiment
run the real experiment, then solve a problem about it

3 build & name
build the machine, then name its parts

4 solve
use your machine to help someone

5 show
show it and say what you learned

At the end – the **level-up build**: one machine that puts two simple machines together, the check you're ready for the next level.



meet the see-saw



In the park, Arpit and his friend love the see-saw. But Arpit is bigger – his side sank and stayed down, leaving his small friend stuck up in the air. They wondered how to make the see-saw fair for them both.

words to know

lever

fulcrum

balance

 **your turn:** tick where you might find a see-saw:

- ☐ a park
- ☐ a kitchen sink
- ☐ a playground



observing & understanding

outcome 1 – you're learning to: measure what the machine does



do the experiment

Balance 0.5kg on the left with 0.5kg on the right. Now add another 0.5kg to the left, so the left holds 1kg.

now work it out: How much must the RIGHT side hold to balance it again?



work it out and write your answer:

my answer ...



show your working here



building & making



Build your see-saw from your kit and model cards. Make it so it **tips up and down and can balance**.



tick when it works

- 1 **Draw** your see-saw in the box below.
- 2 **Label** these two parts on your drawing:

your labels

the plank

the fulcrum

 draw your see-saw here – then label its parts



problem solving

outcome 1 – you're learning to: find what isn't working

Your little brother is too light to lift you on the see-saw. Make it fair so you can both play.



Draw your fix – what did you change to make it work?



draw your fix here



did you know?

A see-saw, scissors and a spoon all work as levers – a bar that turns on one point.



presenting & explaining

show your see-saw to a friend. say three things:

1. **what it is** – “this is my see-saw. it uses a lever.”
2. **its parts** – point to the plank and the fulcrum.
3. **what it does** – “it tips up and down and can balance.”

 the new words i learned – write what each one means:

lever

fulcrum

balance



meet the weighing scale



At the market, Nadia watched the fruit-seller weigh mangoes on a two-pan scale. When both pans held the same, the arm sat perfectly level. 'That's how she knows it's a fair amount,' Nadia thought.

words to know

lever

pan

balance

 **your turn:** tick where you might see a two-pan scale:

- ☐ a fruit market
- ☐ a swimming pool
- ☐ a jeweller's shop



observing & understanding

outcome 1 – you're learning to: measure what the machine does



do the experiment

The left pan holds a 2kg weight. You add equal blocks to the right pan until it sits level – it takes 4 blocks.

now work it out: How heavy is ONE block?



work it out and write your answer:

my answer ...



show your working here



building & making



Build your weighing scale from your kit and model cards. Make it so it **tips, then sits level** when both pans match.



tick when it works

- 1 **Draw** your weighing scale in the box below.
- 2 **Match** each part to what it does – draw a line.

the arm

hold the things you weigh

the two pans

the bar that tips and balances

 now draw your weighing scale



problem solving

outcome 1 – you're learning to: find what isn't working

Help the fruit-seller share fairly – make both pans of the scale sit level.



Draw your fix – what did you change to make it work?

 draw your fix here



did you know?

Every shop scale works this way – it only reads right when it balances.



presenting & explaining

show your weighing scale to a friend. say three things:

1. **what it is** – “this is my weighing scale. it uses a lever.”
2. **its parts** – point to the arm and the two pans.
3. **what it does** – “it tips, then sits level when both pans match.”

 the new words i learned – write what each one means:

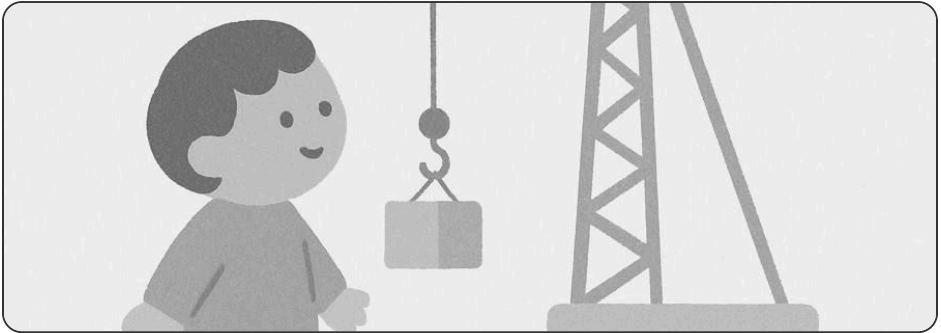
lever

pan

balance



meet the crane



Sara saw a tall crane lift heavy beams to the top of a new building. The worker pulled a rope down low – and the load went up high. How can pulling down lift something up?

words to know

pulley

rope

lift

 **your turn:** tick where you might see a crane:

- ☐ a building site
- ☐ a bathroom
- ☐ a busy port



observing & understanding

outcome 2 – you're learning to: record the pattern in numbers



do the experiment

Lift 2kg straight up by hand, then lift the same 2kg with a single pulley, reading a spring scale each time.

now work it out: A single pulley mainly changes:



circle your answer:

☐

the direction you pull

☐

how heavy the load feels



building & making



Build your crane from your kit and model cards.
Make it so it **lifts a load when you pull the rope down**.



tick when it works

- 1 **Draw** your crane in the box below.
- 2 **Fill in each part** – write it from the word bank.

the rope runs over it to lift ...

the _____

holds the load ...

the _____

word bank

the wheel up high

the hook

 now draw your crane



problem solving

outcome 2 – you're learning to: predict where the fault is

A kitten's food bowl must go up to a high window ledge. Lift it up with your crane.



Draw your fix – what did you change to make it work?

 draw your fix here



did you know?

Flagpoles, well buckets and window blinds all lift with a pulley – pull down, it goes up.



presenting & explaining

show your crane to a friend. say three things:

1. **what it is** – “this is my crane. it uses a pulley.”
2. **its parts** – point to the wheel up high and the hook.
3. **what it does** – “it lifts a load when you pull the rope down.”

 the new words i learned – write what each one means:

pulley

rope

lift



meet the wind turbine



On a windy hill, Kabir watched the great white blades spin round and round. His teacher said the spinning could even lift a load up the tower, with no one climbing at all.

words to know

pulley

wind

lift

 **your turn:** tick where you might see wind turbines:

- ☐ a windy hill
- ☐ inside a fridge
- ☐ out at sea



observing & understanding

outcome 2 – you're learning to: record the pattern in numbers



do the experiment

The blades turn the rope wheel. Each full spin of the wheel winds up 10cm of rope.

now work it out: How far does the load rise in 3 full spins?



work it out and write your answer:

my answer ...



show your working here



building & making



Build your wind turbine from your kit and model cards. Make it so it **spins in the wind and winds a load up**.

☐

tick when it works

1 Draw your wind turbine in the box below.

2 Label these two parts on your drawing:

your labels

the blades

the rope wheel

 draw your wind turbine here – then label its parts



problem solving

outcome 2 – you're learning to: predict where the fault is

A toolbag must reach the top of the tower, with no one climbing. Send it up with your turbine.



Draw your fix – what did you change to make it work?

 draw your fix here



did you know?

Cranes and lifts use pulleys the same way, so no one has to climb to lift a load.



presenting & explaining

show your wind turbine to a friend. say three things:

1. **what it is** – “this is my wind turbine. it uses a pulley.”
2. **its parts** – point to the blades and the rope wheel.
3. **what it does** – “it spins in the wind and winds a load up.”

 the new words i learned – write what each one means:

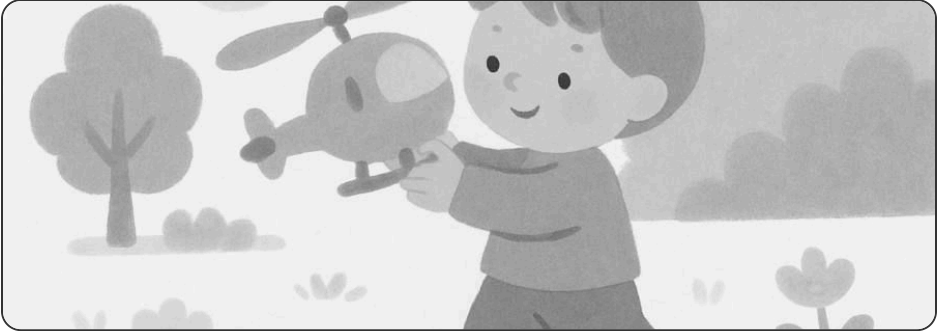
pulley

wind

lift



meet the copter



Mei's toy helicopter has a rotor that spins fast on top. When she turned the little handle slowly, the rotor spun quickly. How did a slow turn become a fast spin?

words to know

gear

teeth

turn



your turn: tick what spins like a copter's rotor:

☐

a fan

☐

a book

☐

a windmill



observing & understanding

outcome 2 – you're learning to: record the pattern in numbers



do the experiment

Chain a small gear (10 teeth) to a big gear (20 teeth). Turn the small gear 8 times.

now work it out: How many times does the BIG gear turn?



work it out and write your answer:

my answer ...



show your working here



building & making



Build your copter from your kit and model cards. Make it so it **turns a slow handle into a fast spin**.



tick when it works

- 1 **Draw** your copter in the box below.
- 2 **Match** each part to what it does – draw a line.

the big gear

spins the rotor fast

the small gear

the handle turns this one



now draw your copter



problem solving

outcome 2 – you're learning to: predict where the fault is

A lamb is lost far across the field. Fly your copter out to look for it.



Draw your fix – what did you change to make it work?



draw your fix here



did you know?

Bicycles, clocks and egg-beaters all use gears to change how fast things spin.



presenting & explaining

show your copter to a friend. say three things:

1. **what it is** – “this is my copter. it uses gears.”
2. **its parts** – point to the big gear and the small gear.
3. **what it does** – “it turns a slow handle into a fast spin.”

 the new words i learned – write what each one means:

gear

teeth

turn

name: _____

openhouse

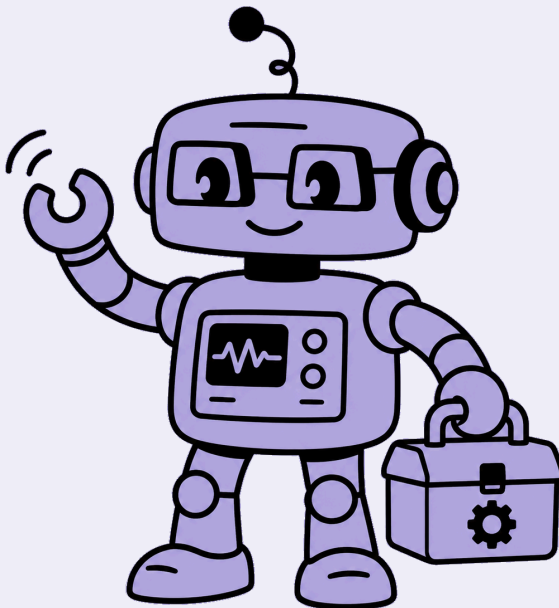
robotics · level 1

◆ mechanics ◆

your portfolio of the openhouse experience

ages 8-12

part 2 · machines 6-10



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the skills you build

You grow all four skills on **every machine**. Each grows in four **outcomes** (the steps below) – the machines climb them in order, a little further each time. You climbed outcomes 1–2 in part 1; this book grows outcomes 3–4.



observing & understanding

1. measure what the machine does →
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 3. compare and work out the rule →
 4. **predict a value, then prove it**
-



building & making

1. build it and read the parts →
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-



problem solving

1. find what isn't working →
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presenting & explaining

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use your machine to help someone

5 show
show it and say what you learned

At the end – the **level-up build**: one machine that puts two simple machines together, the check you're ready for the next level.



meet the bulldozer



A bulldozer pushed a huge pile of earth off the road – slow, but so strong it never got stuck. Ravi watched all that weight move and wondered what made its push so powerful.

words to know

gear

push

slow

 **your turn:** tick what a bulldozer does:

- ☐ push heavy earth
- ☐ tell the time
- ☐ clear a blocked road



observing & understanding

outcome 3 – you're learning to: compare and work out the rule



do the experiment

Chain a small gear (10 teeth) to a big gear (30 teeth). Turn the small gear 9 times.

now work it out: How many times does the BIG gear turn?



work it out and write your answer:

my answer ...



show your working here



building & making



Build your bulldozer from your kit and model cards. Make it so it **gives a slow, strong push**.

☐ tick when it works

- 1 **Draw** your bulldozer in the box below.
- 2 **Fill in each part** – write it from the word bank.

the motor turns it fast ...

the _____

turns slow with a strong push ...

the _____

word bank

the small gear

the big gear

 now draw your bulldozer



problem solving

outcome 3 – you're learning to: change one thing and measure the effect

The school path is blocked with a heavy pile of sand. Clear it with your bulldozer.



Draw your fix – what did you change to make it work?

 draw your fix here



did you know?

Cars and hand drills use gears the strong way – slow, but powerful enough not to stall.



presenting & explaining

show your bulldozer to a friend. say three things:

1. **what it is** – “this is my bulldozer. it uses gears.”
2. **its parts** – point to the small gear and the big gear.
3. **what it does** – “it gives a slow, strong push.”

 the new words i learned – write what each one means:

gear

push

slow



meet the rickshaw



Dadi's shopping was heavy to carry. Ishaan loaded it onto his rickshaw cart and it rolled along easily on its wheels. Carrying was hard – but rolling was easy.

words to know

wheel

axle

roll



your turn: tick what rolls on wheels and an axle:

☐

a cart

☐

a cloud

☐

a car



observing & understanding

outcome 3 – you're learning to: compare and work out the rule



do the experiment

Push your cart the same each time: with 0.5kg it rolled 80cm; with 1 kg it rolled 50cm.

now work it out: With 1.5kg, will it roll MORE or LESS than 50cm?



circle your answer:

☐

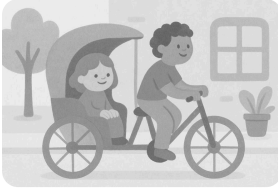
less than 50cm

☐

more than 50cm



building & making



Build your rickshaw from your kit and model cards. Make it so it **rolls a heavy load along easily**.

☐ tick when it works

- 1 **Draw** your rickshaw in the box below.
- 2 **Label** these two parts on your drawing:

your labels

the wheels

the axle

 draw your rickshaw here – then label its parts



problem solving

outcome 3 – you're learning to: change one thing and measure the effect

The fruit-seller's boxes are too heavy to carry to market. Roll them there in your rickshaw.



Draw your fix – what did you change to make it work?

 draw your fix here



did you know?

Cars, trolleys and skateboards all roll on a wheel and axle, so heavy things move easily.



presenting & explaining

show your rickshaw to a friend. say three things:

1. **what it is** – “this is my rickshaw. it uses a wheel and axle.”
2. **its parts** – point to the wheels and the axle.
3. **what it does** – “it rolls a heavy load along easily.”

 the new words i learned – write what each one means:

wheel

axle

roll



meet the fishing rod



By the lake, Mr. Rao swung his long rod up to lift the fish clear of the water, then wound the reel to bring the line in. One tool did two different jobs.

words to know

lever

reel

wind

 **your turn:** tick what a fishing rod does:

- ☐ lift the fish up
- ☐ bake bread
- ☐ wind the line in



observing & understanding

outcome 3 – you're learning to: compare and work out the rule



do the experiment

Swing the long rod up to lift, then wind the reel to bring the line in.

now work it out: A fishing rod is TWO simple machines in one. Which two?



work it out and write your answer:

my answer ...



show your working here



building & making



Build your fishing rod from your kit and model cards. Make it so it **swings up to lift**, then **winds the line in**.

☐ tick when it works

- 1 **Draw** your fishing rod in the box below.
- 2 **Match** each part to what it does – draw a line.

the long rod

winds the line in

the reel

swings up to lift, like a lever

 now draw your fishing rod



problem solving

outcome 3 – you're learning to: change one thing and measure the effect

Mr. Rao's fishing net is too heavy to lift by hand. Swing it up, then wind it in.



Draw your fix – what did you change to make it work?

 draw your fix here



did you know?

A crane with a winch does the same two jobs – lift with an arm, wind with a reel.



presenting & explaining

show your fishing rod to a friend. say three things:

1. **what it is** – “this is my fishing rod. it uses two machines in one.”
2. **its parts** – point to the long rod and the reel.
3. **what it does** – “it swings up to lift, then winds the line in.”

 the new words i learned – write what each one means:

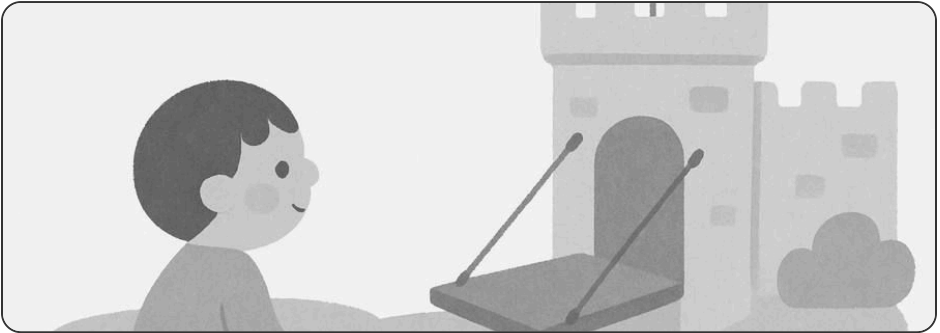
lever

reel

wind



meet the drawbridge



A tall sailboat needed to pass under the bridge. The bridge-keeper pulled a rope and the bridge deck lifted up to let it through – then lowered again for the cars.

words to know

lever

pulley

raise

 **your turn:** tick when a drawbridge lifts:

- ☐ to let a tall boat pass
- ☐ to make tea
- ☐ then lowers for cars



observing & understanding

outcome 4 – you're learning to: predict a value, then prove it



do the experiment

The rope wheel winds up 8cm of rope for every turn you make.

now work it out: How many turns raise the bridge deck by 24cm?



work it out and write your answer:

my answer ...



show your working here



building & making



Build your drawbridge from your kit and model cards. Make it so it **raises its deck when you pull the rope**.

☐ tick when it works

- 1 **Draw** your drawbridge in the box below.
- 2 **Fill in each part** – write it from the word bank.

the road that lifts up ...

the _____

pull its rope to raise the deck ...

the _____

word bank

the deck

the rope wheel

 now draw your drawbridge



problem solving

outcome 4 – you're learning to: solve it and explain why it worked

A tall boat needs to pass, but the bridge is down. Raise your drawbridge to let it through.



Draw your fix – what did you change to make it work?

 draw your fix here



did you know?

Lifting river bridges and castle gates raise the same way – a rope over a wheel.



presenting & explaining

show your drawbridge to a friend. say three things:

1. **what it is** – “this is my drawbridge. it uses two machines in one.”
2. **its parts** – point to the deck and the rope wheel.
3. **what it does** – “it raises its deck when you pull the rope.”

 the new words i learned – write what each one means:

lever

pulley

raise



meet the tow truck



A car was stuck deep in the mud after the rain. The tow truck rolled up close, then its winch wound the rope and hauled the car out to safety.

words to know

wheel

winch

haul

 **your turn:** tick what a tow truck does:

- ☐ roll up close
- ☐ cook dinner
- ☐ winch the car in



observing & understanding

outcome 4 – you're learning to: predict a value, then prove it



do the experiment

Roll the truck to the car (wheels), then wind the winch to haul it in (a winch is a reel + rope).

now work it out: A tow truck is TWO simple machines in one. Which two?



work it out and write your answer:

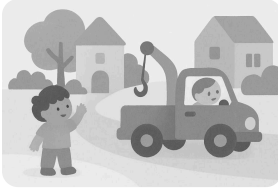
my answer ...



show your working here



building & making



Build your tow truck from your kit and model cards. Make it so it **rolls up close, then winches the load in.**

☐ tick when it works

- 1 **Draw** your tow truck in the box below.
- 2 **Label** these two parts on your drawing:

your labels

the wheels

the winding reel

 draw your tow truck here – then label its parts



problem solving

outcome 4 – you're learning to: solve it and explain why it worked

A cart is stuck fast in the mud. Haul it out to safety with your tow truck.



Draw your fix – what did you change to make it work?

 draw your fix here



did you know?

Cranes and cable cars use a winch to haul heavy loads that wheels alone can't move.



presenting & explaining

show your tow truck to a friend. say three things:

1. **what it is** – “this is my tow truck. it uses two machines in one.”
2. **its parts** – point to the wheels and the winding reel.
3. **what it does** – “it rolls up close, then winches the load in.”

 the new words i learned – write what each one means:

wheel

winch

haul



your level-up build

This is your **level check**. Build **one machine that puts two simple machines together** – a lever and a pulley, gears and wheels, your choice. Show all four skills on it.

✓ you are ready for the next level when you can:

- ☐ predict a value, then prove it
- ☐ improve it and measure the change
- ☐ solve it and explain why it worked
- ☐ present it and answer a question

draw your level-up build here